

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C, C++, Assembly, Python, Java

Embedded & Firmware: TTC 60 VCU, STM32H7, PX4, real-time firmware, fixed-point arithmetic, deterministic execution, CAN, SPI, microcontroller I/O

Controls & Algorithms: Fixed-point PI/PID, anti-windup, saturation, slew-rate limiting, state estimation, sensor fusion

Test & Tools: Software-in-the-loop replay, regression tests, pytest, GDB, TASKING, TTC Downloader, BusMaster, MoTeC i2 Pro, Git

EXPERIENCE

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning for VCU controls, embedded firmware, telemetry, CAN diagnostics, and validation workflows
- Coordinate priorities and technical reviews across power limiting, launch control, dashboard, simulation, and vehicle testing

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Developed and validated embedded flight-control firmware in C on PX4, iterating control gains against logged flight telemetry
- Integrated and tuned visual-inertial odometry for onboard state estimation, improving position-hold accuracy
- Tuned PID flight-control loops to reduce vertical and horizontal position drift in autonomous flight

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Implemented a torque-to-power controller in C with fixed-point PI control, saturation, anti-windup, and slew limits for deterministic TTC 60 VCU execution
- Built CAN diagnostics firmware for live power and torque frames, status bits, and error counters, debugging bus behavior against real-time BusMaster captures
- Designed a lap-adaptive energy-management algorithm setting a power setpoint from cumulative energy versus budget, holding actual power within **0.01%** of setpoint

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Built deterministic entry and exit state logic for power limiting from measured power, requested torque, and threshold flags
- Tuned the PI torque controller used for power limiting to within **2%** error, smoothing torque transitions to preserve drivability
- **Results:** 1st in endurance at MIS EV 2025

PROJECT EXPERIENCE

Formula SAE EV Dashboard

Feb 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

C, STM32H7, CAN, SPI

- Built a real-time CAN telemetry display on an STM32H7 driving an LCD over SPI, decoding HV voltage, throttle, cell temperature, and power budget
- Designed an energy bar and LED alert system giving the driver lap-by-lap feedback on energy budget

SR-Wireless-CAN — Wireless Firmware Flashing & Telemetry

2026

Spartan Racing, Formula SAE | San Jose, CA | Backend lead

Python, python-can, CAN trace analysis, FastAPI, WebSockets, SQLite, Docker

- Proposed and implemented wireless firmware flashing for the TTC 60 VCU, decoding the undocumented flashing sequence from CAN trace captures and reimplementing it in Python over the bus
- Led backend development for the telemetry platform, streaming live CAN signals to a browser dashboard with DBC-driven decoding and recorded-trace replay